CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 4: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A parallelogram is a rhombus if its diagonals: 1

(a) are perpendicular **(b)** are in the ratio 1:2
(c) are equal in length **(d)** make an angle of 60°

2. The angles of a quadrilateral are x°, $(x+6)^\circ$, $(x+12)^\circ$ and $(x+18)^\circ$. The value of x is: 1

(a) 91 **(b)** 87 **(c)** 81 **(d)** 75

3. The adjacent sides of a parallelogram are in the ratio 5:6 and its perimeter is 110 cm. The longer side is: 1

(a) 55 cm **(b)** 60 cm **(c)** 30 cm **(d)** 25 cm

4. The diagonals of a parallelogram ABCD meet at O. If AO $= (5x-3)$ cm and OC $= (3x+7)$ cm, then AC is: 1

(a) 44 cm **(b)** 22 cm **(c)** 46 cm **(d)** 5 cm

5. In a rhombus ABCD, $\angle A = (5x+20)^\circ$ and $\angle B = (3x+32)^\circ$. The value of x is: 1

(a) 16 **(b)** 20 **(c)** 17 **(d)** 15

6. In a kite ABCD with AB = AD and CB = CD, $\angle A = 50^\circ$ and $\angle C = 120^\circ$. The angle $\angle B$ is: 1

(a) 190° **(b)** 85° **(c)** 95° **(d)** 105°

7. Three exterior angles of a quadrilateral are 60°, 125° and 90°. The fourth exterior angle is: 1

(a) 95° **(b)** 85° **(c)** 110° **(d)** 75°

8. Which statement is true? 1

(a) Every kite is a rhombus **(b)** Every parallelogram is a rectangle
(c) Every rhombus is a parallelogram **(d)** Every trapezium is a parallelogram

9. The angles of a quadrilateral are x°, $(x+14)^\circ$, $(x+28)^\circ$ and $(x+42)^\circ$. The value of x is: 1

(a) 69 **(b)** 55 **(c)** 79 **(d)** 83

10. The adjacent sides of a parallelogram are in the ratio 1:4 and its perimeter is 90 cm. The longer side is: 1

(a) 45 cm **(b)** 9 cm **(c)** 36 cm **(d)** 72 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The exterior angles at three vertices of a quadrilateral are $65°$, $100°$ and $88°$. Find the fourth exterior angle and the fourth interior angle. 2

12. The angles of a quadrilateral are in the ratio 2:5:7:6. Find the largest angle. 2

13. The diagonals of a parallelogram ABCD meet at O with AO $= (4x + 6)$ cm and OC $= (3x + 10)$ cm. Find x and AC. 2

14. The exterior angles at three vertices of a quadrilateral are $110°$, $84°$ and $90°$. Find the fourth exterior angle and the fourth interior angle. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The angles of a quadrilateral are $x°$, $(x + 24)°$, $(x + 48)°$ and $(x + 72)°$. Find all the angles. 3

16. The adjacent sides of a parallelogram are in the ratio 3:8 and its perimeter is 110 cm. Find the lengths of its sides. 3

17. In a rectangle ABCD, $\angle ACB = 22°$. Find $\angle CAB$, $\angle AOB$ and $\angle ODC$ where O is the point of intersection of the diagonals. 3

OR

The angles of a quadrilateral are $x°$, $(x + 4)°$, $(x + 8)°$ and $(x + 12)°$. Find all the angles. 3

18. The adjacent sides of a parallelogram are in the ratio 6:7 and its perimeter is 104 cm. Find the lengths of its sides. 3

OR

In a rectangle ABCD, $\angle ACB = 28°$. Find $\angle CAB$, $\angle AOB$ and $\angle ODC$ where O is the point of intersection of the diagonals. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. ABCD is a rhombus whose diagonals meet at O. $\angle OAB = (3x)°$ and $\angle OBA = (6x)°$. Find x, $\angle A$ and $\angle B$ of the rhombus. 5

OR

The diagonals of a rectangular frame ABCD are AC $= (7x + 4)$ cm and BD $= (5x + 16)$ cm. Find x, the length of each diagonal, OA (O is the centre) and the length of wire needed to cover both diagonals. 5

20. ABCD is a rhombus whose diagonals meet at O. $\angle OAB = (7x - 3)°$ and $\angle OBA = (2x + 3)°$. Find x, $\angle A$ and $\angle B$ of the rhombus. 5

OR

The diagonals of a rectangular frame ABCD are AC $= (8x + 3)$ cm and BD $= (5x + 18)$ cm. Find x, the length of each diagonal, OA (O is the centre) and the length of wire needed to cover both diagonals. 5

END OF QUESTION PAPER — Total: 40 Marks